

STRATUM CAPITAL BANK

Enterprise Governance & Risk Framework

Chapter 3 — Enterprise AI Governance & Decision Support Framework

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DISTRIBUTION**

*Prerequisites: Chapter 1 (Bank Overview) and Chapter 2 (Lending
Philosophy & Risk Appetite) must be read first.*

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Chapter 3 — Enterprise AI Governance & Decision Support Framework

3.1 Purpose of This Chapter

This chapter defines the comprehensive governance framework applicable to all Artificial Intelligence and Machine Learning systems deployed by Stratum Capital Bank in support of its credit assessment and risk management activities. It establishes the principles, policies, structures, and operational standards that govern how AI systems are developed, validated, approved, deployed, monitored, and retired within the Bank's regulated environment.

The provisions of this chapter extend and operationalise the AI Governance Philosophy introduced in Chapter 1 and the AI-Assisted Decision Philosophy established in Chapter 2. All AI-related policies, operational procedures, technical standards, and vendor agreements across the Bank must be consistent with the requirements of this chapter. In the event of conflict between a subordinate AI policy document and this chapter, the provisions of this chapter take precedence.

This chapter applies to all AI systems currently operated by the Bank, all future AI systems proposed for deployment in a credit or risk context, and all staff whose roles involve the operation, supervision, development, or audit of AI systems. The chapter also establishes governance obligations for third-party AI technology providers engaged by the Bank.

Policy Note: *This chapter does not repeat the organisational structure, risk appetite, or lending decision philosophy established in Chapters 1 and 2. Readers must be familiar with those chapters before applying the provisions herein.*

3.2 AI Vision

Stratum Capital Bank's AI vision is to operate as an institution in which Artificial Intelligence amplifies the analytical capability of qualified banking professionals without diminishing the primacy of human judgment, the transparency of decision-making, or the fairness of outcomes for all borrowers.

The Bank does not pursue AI deployment as an end in itself. AI is adopted where it demonstrably improves the quality, consistency, and efficiency of credit assessment and risk management — and only where the Bank can maintain comprehensive governance, explainability, and human oversight over the AI system's operation and outputs.

The Bank's AI vision rests on four pillars:

- **Trustworthy AI** — AI systems operate reliably, predictably, and consistently within defined parameters, producing outputs that Credit Officers can interpret, interrogate, and if appropriate, override.
- **Explainable AI** — Every AI output presented to a Credit Officer is accompanied by a plain-language explanation of its derivation. The Bank does not deploy black-box models in credit-related decision support.
- **Ethical AI** — AI systems are designed, validated, and monitored to ensure they do not produce discriminatory outcomes, perpetuate historical lending bias, or violate any applicable non-discrimination principle.

- **Governed AI** — Every AI system operates under formal lifecycle governance, documented in the Model Inventory Register, subject to independent validation, and reviewed on a scheduled and event-driven basis.

3.3 AI Governance Principles

The following ten principles govern all AI-related activity within Stratum Capital Bank. They are non-negotiable. No commercial objective, operational efficiency target, or technology roadmap consideration justifies departure from these principles:

- **Human Supremacy** — Human authority is always superior to AI output. No AI recommendation, score, or classification constitutes a credit decision. Final authority rests exclusively with authorised human officers.
- **Explainability Obligation** — Every AI output used in a credit context must be explainable to the reviewing officer in terms that allow meaningful professional evaluation. Unexplainable outputs shall not be used in credit decision support.
- **Proportionality** — The level of AI governance applied to a system is proportionate to the risk that system poses if it performs incorrectly, is misused, or produces biased outputs.
- **Independence of Validation** — AI models must be validated by a function independent of the team that developed them. Self-validation is prohibited.
- **Transparency of Limitations** — All known limitations, assumptions, and performance boundaries of deployed AI systems must be documented, communicated to users, and reflected in governance controls.
- **Data Integrity** — AI models are only as reliable as the data used to train and operate them. Data quality, lineage, and access controls are treated as first-order governance obligations.
- **Non-Discrimination** — AI systems must not produce or reinforce discriminatory outcomes in credit assessment. Fairness audits are a mandatory component of the model lifecycle.
- **Audit Completeness** — Every AI-assisted credit decision must generate a complete, tamper-proof audit trail, retained in accordance with the Bank's data governance standards.
- **Continuous Improvement** — AI governance is not static. Model performance, fairness, and relevance are continuously monitored, and governance standards are updated to reflect technological, regulatory, and institutional learning.
- **Accountability** — Every AI system has a designated Owner within the AI Governance and Analytics Department. Every governance obligation has a designated Responsible Party. Diffuse or unclear accountability is a governance failure.

3.4 AI Systems Currently Operated by the Bank

Stratum Capital Bank currently operates the following AI systems in support of its credit assessment and risk management activities. Each system is registered in the Model Inventory Register maintained by the AI Governance and Analytics Department:

AI System	Primary Function	Output Type	Human Review Required
Credit Risk Prediction Model	Estimate Probability of Default (PD)	Numeric PD%; Risk Tier	Yes — mandatory at all tiers
Explainable AI (XAI) Module	Explain PD drivers in plain language	Factor ranking; directional influence	Yes — reviewed with PD output
RAG Policy Retrieval Engine	Retrieve relevant internal policy sections	Policy text excerpts; citations	Yes — officer validates applicability
Internal Policy Search Engine	Full-text search across policy library	Ranked document matches	Yes — officer interprets results
Decision Support Dashboard	Consolidate AI outputs for officer review	Unified view; audit trail	Yes — primary officer interface

Policy Note: No AI system listed above constitutes a decision-making authority. Each system produces outputs that inform, not determine, the professional judgment of authorised Credit Officers. The Decision Support Dashboard does not commit the Bank to any credit position; it is an information and audit interface only.

3.5 Human-in-the-Loop Decision Framework

The Human-in-the-Loop (HITL) Decision Framework, introduced in Chapter 1 and applied to credit risk tiers in Chapter 2, is restated here in its AI governance context to define specifically how human oversight obligations interact with AI system outputs across the credit assessment lifecycle.

3.5.1 HITL Requirements by Risk Tier

The Enterprise Decision Matrix below governs the minimum human review requirements for each AI risk tier classification. These requirements apply to every application processed through the Bank's AI Decision Support System, without exception:

Risk Tier	PD Range	AI Recommendation	Minimum Human Action	Commitment Authority
Tier 1 — Low	0% – 35%	Recommend Approval	Loan Officer: review checklist + AI output	Credit Manager
Tier 2 — Medium	>35% – 50%	Manual Review	Senior Credit Analyst: assessment memorandum	Credit Manager
Tier 3 — High	>50%	Recommend Rejection	Credit Manager: full file review; exception path if applicable	Chief Credit Officer (exception only)

3.5.2 HITL Integrity Obligations

The HITL framework imposes the following non-negotiable integrity obligations on all staff interacting with AI outputs:

- No Credit Officer may submit an application for commitment without completing the AI output review step as defined for their role and the applicable risk tier.
- No AI output may be communicated to an applicant as a final institutional decision without prior authorisation by a qualified human officer holding the appropriate Delegation of Authority.
- AI outputs must be reviewed in conjunction with the full application file. Review of the AI output in isolation — without reference to underlying documentation — does not satisfy the HITL requirement.
- The review of AI output must be documented. Undocumented reviews are treated as non-compliant with HITL requirements and subject to escalation to the Chief Credit Officer.

Example — HITL Compliance Failure: *A Loan Officer receives a Tier 1 (Low Risk) PD classification for a personal loan application and, under time pressure, notifies the applicant of approval without completing the Standard Credit Review Checklist or forwarding the file to the Credit Manager for commitment authority. This constitutes a material HITL breach. The approval is invalid under this Framework. The matter is escalated to the Credit Manager and recorded in the Compliance Incident Register.*

3.6 AI Decision Lifecycle

The AI Decision Lifecycle defines the sequence of stages through which an individual credit application moves within the Bank's AI Decision Support System, from initial submission to audit log finalisation. Understanding the lifecycle is a training requirement for all Credit Officers, Senior Credit Analysts, and Credit Managers.

3.6.1 Lifecycle Stages

- Stage 1 — Application Submission: The Loan Officer submits a complete credit application, including all required documentation, through the Bank's credit origination system. Incomplete applications are returned to the officer. Submission triggers the AI Decision Support System.
- Stage 2 — Data Ingestion and Preprocessing: The system ingests structured application data and performs automated data quality checks. Applications with critical data quality failures are flagged and returned to the Loan Officer for resolution before AI assessment proceeds.
- Stage 3 — PD Model Computation: The Credit Risk Prediction Model computes a Probability of Default percentage based on the application's input variables and the model's trained parameters. This computation is logged in real time in the Decision Audit Register.
- Stage 4 — Risk Tier Assignment: The PD output is mapped to a risk tier (Low, Medium, or High) based on the thresholds defined in Chapter 2 and this chapter. The tier assignment determines the mandatory process pathway.
- Stage 5 — XAI Explanation Generation: The Explainable AI Module generates a plain-language summary of the primary factors contributing to the PD estimate, including each factor's directional influence. This summary is presented to the reviewing officer alongside the PD score.
- Stage 6 — Policy Retrieval: The RAG Policy Retrieval Engine identifies and retrieves the most relevant internal policy sections applicable to the application type and risk tier. Retrieved policies are displayed in the Decision Support Dashboard for officer reference.

- Stage 7 — Officer Review: The assigned Credit Officer reviews the complete Decision Support Dashboard output — PD score, risk tier, XAI summary, retrieved policy references, and full application file — and exercises professional judgment to form a preliminary assessment.
- Stage 8 — Human Decision and Documentation: The authorised officer makes and documents a credit decision or recommendation in accordance with the HITL requirements for the applicable risk tier. The decision is recorded in the credit file and the Decision Audit Register.
- Stage 9 — Audit Log Finalisation: The Decision Audit Register is updated with all relevant decision data, including AI outputs, officer identity, decision outcome, override notation (if applicable), and timestamp. The audit record is locked and cannot be modified without a formal audit amendment process.

Policy Note: No stage in the AI Decision Lifecycle constitutes an autonomous credit decision. Stages 1 through 6 are AI-facilitated information processing steps. Stage 7 is the essential human judgment step. Stage 8 is the human decision step. Stage 9 is the governance record step. Stages 7, 8, and 9 cannot be automated, delegated to AI, or bypassed.

3.7 Explainable AI Policy

Explainability is a non-negotiable governance requirement for all AI systems deployed by Stratum Capital Bank in a credit context. The Bank does not deploy opaque or black-box models in credit decision support, irrespective of their predictive performance. A model that cannot explain its outputs in terms accessible to a qualified Credit Officer does not meet the Bank's deployment standards.

3.7.1 XAI Minimum Standards

All AI systems producing credit-relevant outputs must satisfy the following explainability standards:

- **Factor Identification** — The system must identify the primary variables that contributed most significantly to the PD estimate for each individual application. A minimum of five primary factors must be identified and ranked in order of contribution magnitude.
- **Directional Clarity** — For each identified factor, the system must indicate whether its contribution is risk-increasing (raising PD) or risk-mitigating (reducing PD), and provide an accessible description of the factor's relevance.
- **Plain Language Output** — Explanations must be written in language comprehensible to a Credit Officer without specialist data science training. Technical notation without plain-language equivalents is not acceptable.
- **Individual-Level Explanation** — Explanations must be generated at the individual application level. Global model explanations that describe general model behaviour are supplementary and do not substitute for individual-case explanations.
- **Consistency and Reproducibility** — The same application inputs must produce the same XAI explanation upon repeated processing. Stochastic variation in explanations for identical inputs is a model quality failure requiring investigation.

3.7.2 Officer Obligations Regarding XAI Output

Credit Officers are trained to evaluate XAI explanations critically. Officers must:

- Read the full XAI explanation, not only the PD percentage and risk tier.
- Assess whether the identified primary factors are consistent with the qualitative information in the application file.
- Identify any material risk factors visible in the application file that are not reflected in the XAI explanation, and document these in the credit assessment memorandum or review checklist.
- Flag any XAI explanation that appears inconsistent, internally contradictory, or implausible to the AI Governance and Analytics Department for investigation.

Policy Note: An XAI explanation that officers cannot interpret, that attributes significant risk weight to variables the officer cannot validate in the application file, or that is inconsistent with the borrower's observable financial position must be treated as a system quality concern — not as a reason to accept the model output uncritically.

3.8 Model Transparency

Stratum Capital Bank maintains comprehensive documentation of all deployed AI models in the Model Inventory Register. Transparency is a governance obligation — not a desirable attribute. Officers who interact with AI outputs are entitled to understand the nature and limitations of the models they are using.

Model Transparency documentation maintained in the Model Inventory Register includes:

- Model purpose and scope — The specific credit assessment context the model is designed to address, and the borrower population and product types for which the model was trained and validated.
- Training data description — The sources, time period, volume, and key characteristics of the data used to train the model, including any known sampling limitations or exclusions.
- Input variable list — A complete list of variables used as model inputs, with descriptions of how each variable is sourced and any known data quality risks.
- Model architecture summary — A non-technical description of the model's analytical approach, sufficient for a governance audience to understand the model's general methodology without requiring data science expertise.
- Known limitations — A frank documentation of the model's known limitations, including the borrower populations or economic conditions under which the model's performance may be less reliable.
- Performance history — A record of the model's performance metrics over time, including any deterioration events, remediation actions taken, and retraining history.

Model Transparency documentation is available to Credit Officers on request through the AI Governance and Analytics Department. Summary model cards — concise non-technical overviews — are made available on the Bank's internal knowledge platform for all staff holding credit authority.

3.9 Bias Detection and Fairness Policy

Stratum Capital Bank treats algorithmic bias as a material risk — not merely a technical concern. Biased model outputs can produce discriminatory lending outcomes, create regulatory exposure, and damage the Bank's reputation and community trust. Bias detection and fairness assessment are mandatory components of the model lifecycle at initial deployment and on a continuous basis thereafter.

3.9.1 Fairness Assessment Framework

Fairness Dimension	Assessment Method	Acceptable Threshold	Breach Action
Approval Rate Disparity	Demographic parity analysis across borrower segments	< 5% unexplained differential	Immediate CRO escalation; model review
PD Score Distribution	Score distribution comparison across segments	No statistically significant adverse skew	Model audit; feature review
Feature Proxy Detection	Correlation analysis of input variables with protected characteristics	Zero tolerance for direct proxies	Feature removal; re-validation
XAI Explanation Consistency	Explanation coherence testing across demographic groups	Explanation variance < 10%	XAI module audit
Override Rate by Segment	Officer override analysis by applicant segment	No systematic directional pattern	Compliance + CRO review

3.9.2 Bias Remediation

Where a fairness assessment identifies a bias finding that meets or exceeds the defined breach threshold, the following remediation sequence is mandatory:

- Immediate notification of the CRO and Chief Compliance Officer, with a written Bias Finding Report completed within 24 hours of identification.
- Interim enhanced monitoring of credit decisions in the affected demographic segment while the model is under review.
- Root cause analysis conducted by the AI Governance and Analytics Department within five business days, identifying the source of the disparate impact — whether in training data, feature selection, or model architecture.
- Remediation action — which may include feature removal, data rebalancing, model recalibration, or model replacement — implemented in accordance with the Model Validation requirements of Section 3.10.
- Post-remediation fairness assessment confirming resolution before the remediated model is returned to production.

Policy Note: *The Bank's bias detection programme examines outcomes, not intentions. A model that produces disparate impacts without intentional discriminatory design still requires remediation. Intent is not a defence against disproportionate outcome.*

3.10 Model Validation Policy

All AI models used in a credit context must undergo formal validation before initial deployment and upon any material change, recalibration, or retraining. Validation is conducted by the Bank's independent Model Validation Unit, which reports to the Chief Risk Officer and operates separately from the AI Governance and Analytics Department responsible for model development.

3.10.1 Validation Scope

The scope of each model validation exercise includes:

- Conceptual soundness review — Assessment of whether the model's design and methodology are theoretically appropriate for the credit risk estimation purpose it is intended to serve.
- Data quality and representativeness review — Examination of the training data for completeness, accuracy, recency, and representativeness of the Bank's current borrower population.
- Performance benchmarking — Testing of the model's predictive accuracy, calibration, and discriminatory power against the minimum standards defined in Section 3.11 (Model Monitoring).
- Bias and fairness audit — Completion of the full fairness assessment framework defined in Section 3.9.1.
- Stress testing — Assessment of model performance under adverse economic scenarios defined by the Chief Risk Officer.
- XAI consistency check — Verification that the Explainable AI module is producing explanations that are consistent with the underlying model's actual decision logic.
- Documentation review — Confirmation that all Model Transparency documentation required under Section 3.8 is complete, accurate, and current.

3.10.2 Validation Output

The Model Validation Unit produces a formal Validation Report for each model reviewed. The Validation Report is submitted to the Model Governance Committee, which determines whether the model is approved for deployment, approved with conditions, or rejected for further development. Models may not be deployed in a production credit environment without a current, in-scope Validation Report and a formal Model Approval Certificate issued by the Model Governance Committee.

3.11 Model Monitoring Policy

Deployed AI models are subject to continuous performance monitoring. Models approved for production are not static — their performance evolves as economic conditions, borrower behaviour, and the Bank's portfolio composition change over time. Monitoring is the mechanism through which the Bank detects performance drift before it results in material credit quality deterioration.

3.11.1 Performance Monitoring Indicators

Performance Metric	Minimum Standard	Alert Threshold	Review Frequency	Escalation
Gini Coefficient (discriminatory power)	≥ 0.45	< 0.50	Quarterly	CRO
Brier Score (calibration accuracy)	≤ 0.20	> 0.18	Quarterly	CRO

Performance Metric	Minimum Standard	Alert Threshold	Review Frequency	Escalation
Population Stability Index (PSI)	< 0.10	> 0.08	Monthly	AI Governance
Default Prediction Accuracy (Tier alignment)	≥ 85%	< 88%	Quarterly	CRO
AI Override Rate (officer divergence)	< 15% of cases	> 12%	Monthly	CCO + CRO
Model Response Time (dashboard latency)	≤ 3 seconds	> 2.5 seconds	Weekly	Technology

3.11.2 Monitoring Governance

The AI Governance and Analytics Department produces a monthly AI Model Performance Dashboard, distributed to the Chief Risk Officer, Chief Credit Officer, and the Model Governance Committee. The dashboard summarises all monitored indicators against their standards and alerts. The CRO reviews the dashboard and determines whether any findings require immediate escalation, formal investigation, or scheduled retraining.

Quarterly, the Model Governance Committee convenes to review consolidated model performance across all deployed systems, assess trends, and determine whether scheduled retraining is warranted based on performance trajectory rather than threshold breach alone.

3.12 Model Retraining Policy

Model retraining is the process of refreshing or recalibrating an AI model's parameters to reflect new data, address identified performance deterioration, or incorporate approved changes to the model's scope or feature set. Retraining is not a routine operational activity — it is a formal governance event subject to the full requirements of the model lifecycle.

3.12.1 Triggers for Retraining

Retraining is initiated when any of the following conditions are met:

- Population Stability Index exceeds 0.10, indicating material drift between the model's training population and its current operational population.
- Gini Coefficient falls below 0.45, indicating deterioration in the model's discriminatory power below the minimum standard.
- Annual scheduled retraining, triggered automatically after twelve months of production operation regardless of performance indicators.
- Material change in the Bank's product portfolio, borrower demographics, or economic environment that the Model Governance Committee determines is likely to reduce model relevance.
- Post-incident retraining required following a Critical or High-severity AI incident finding.

3.12.2 Retraining Process Requirements

All retraining activities must comply with the following process requirements:

- A Retraining Justification Report must be prepared by the AI Governance and Analytics Department, documenting the trigger, the proposed data refresh scope, and any proposed changes to model architecture or features.
- Retraining must be conducted in an isolated development environment. The production model continues to operate during retraining.
- The retrained model must complete the full Validation Process defined in Section 3.10 before any production deployment. Partial validation is not permitted.
- Deployment of a retrained model follows the standard Change Control process, with a cutover plan reviewed by the Technology Department and approved by the CRO.
- Post-deployment monitoring of a retrained model is intensified for the first 90 days, with weekly performance reports replacing the standard monthly cycle.

Policy Note: *Emergency retraining — triggered by a Critical-severity incident — follows an expedited process in which the timeframes above are compressed under CRO direction. However, the validation requirement is not waived in an emergency retraining. If full validation cannot be completed within the required incident resolution SLA, the fallback procedures defined in the Business Continuity section (Section 3.20) are activated while retraining proceeds.*

3.13 AI Incident Management

An AI incident is any event in which an AI system produces outputs that are materially incorrect, evidence bias, are used in a manner inconsistent with this Framework, or result in a credit decision that causes harm to the Bank or a borrower. All staff interacting with AI systems are responsible for reporting suspected AI incidents to the AI Governance and Analytics Department without delay.

3.13.1 Incident Severity and Response Matrix

Severity	Definition	Immediate Action	Notification	Resolution SLA
Critical	Model producing materially incorrect PD outputs; data breach; regulatory exposure	Suspend model; revert to manual assessment	CRO + CEO + Board (within 2 hours)	48 hours
High	Performance degradation beyond alert threshold; significant bias finding	Heightened monitoring; interim controls	CRO + Chief Credit Officer (same day)	5 business days
Medium	Model drift approaching threshold; unexplained override spike	Formal investigation; monitoring increase	CRO (within 1 business day)	10 business days
Low	Minor anomaly; isolated data quality issue; user error	Log; investigate; remediate	AI Governance Lead (next working day)	20 business days

3.13.2 Incident Reporting Obligations

Any member of staff who identifies or suspects an AI incident must submit an AI Incident Report to the AI Governance and Analytics Department within the same business day. Incident Reports must document: the system involved; the nature of the suspected issue; the application(s) affected; the officer who identified the issue; and any interim actions already taken. Failure to report a suspected AI incident in a timely manner is a disciplinary matter.

3.13.3 Post-Incident Review

All Critical and High-severity incidents trigger a mandatory Post-Incident Review conducted by the AI Governance and Analytics Department within 15 business days of incident resolution. The Post-Incident Review produces a written Root Cause Analysis and Remediation Report, which is submitted to the CRO and presented to the Model Governance Committee. Medium and Low-severity incidents are captured in the AI Incident Register and reviewed in aggregate at the quarterly Model Governance Committee meeting.

3.14 Data Governance for AI Systems

The quality, security, and appropriate use of data are foundational to the reliability and fairness of the Bank's AI systems. Data governance standards applicable to AI systems are defined below and must be observed by all staff and third-party providers involved in the collection, processing, storage, or transmission of data used in AI-related activities.

Data Category	Permissible Use	Prohibited Use	Retention Period
Applicant Financial Data	Credit assessment; model training (anonymised); audit	Marketing; third-party sale; non-credit scoring	7 years post facility closure
AI Model Input Variables	PD calculation; XAI explanation; audit log	Profiling outside credit context	Life of model + 5 years
Model Output (PD scores)	Credit file documentation; dashboard display; audit trail	Client communication as final decision; external sharing	10 years
Override Records	Model performance monitoring; credit quality review	Officer performance appraisal as sole metric	10 years
RAG Retrieval Logs	Policy usage analytics; system improvement	Individual officer surveillance	3 years
Audit Logs	Regulatory examination; internal audit; incident investigation	Routine operational monitoring beyond defined scope	10 years

3.14.1 Data Lineage and Provenance

The AI Governance and Analytics Department maintains a Data Lineage Register documenting the source, transformation history, and current location of all datasets used in model training, validation, and production operation. Data whose provenance cannot be fully traced is not permitted for use in model training or production data pipelines.

3.14.2 Data Quality Standards

Input data submitted to the AI Decision Support System must meet defined quality thresholds before the model computation proceeds. Automated data quality checks are applied at Stage 2 of the AI Decision Lifecycle (Section 3.6). Applications failing data quality thresholds are returned to the Loan Officer for correction and are not assigned a provisional AI risk tier on the basis of incomplete or invalid data.

3.15 Data Privacy in AI Operations

Personal and financial data processed by the Bank's AI systems is subject to the Bank's Data Privacy Policy and all applicable data protection principles. The following specific obligations apply in the AI context:

- Minimum Necessary Data — AI models are trained and operated using the minimum data necessary to achieve their stated purpose. Personal data is not incorporated as an AI model input variable unless a clearly documented analytical justification exists.

- Anonymisation for Training — Where AI model training involves historical borrower data, that data must be anonymised or pseudonymised before use in the training environment. Production borrower identities must never be present in model development environments.
- Data Subject Access — Borrowers whose applications are assessed using AI systems are entitled, upon request, to information about the types of data used in that assessment and the categories of factors considered. The Bank satisfies this obligation through its standard Data Subject Access Request procedure.
- Automated Decision Disclosure — Where the Bank's AI system plays a significant role in a credit outcome, the Bank's credit communications must accurately characterise the role of the AI system — as a decision support tool — and confirm that a human officer made the final determination.
- Cross-Border Data Transfer Restrictions — Borrower data processed by AI systems must not be transferred to jurisdictions that do not provide equivalent data protection standards without explicit governance approval from the Chief Risk Officer and Chief Compliance Officer.

Policy Note: *The Bank does not represent AI-generated outputs to borrowers as autonomous institutional decisions. All borrower-facing communications describe credit outcomes as the result of a human credit assessment process supported by analytical tools. The existence and role of AI systems are disclosed in the Bank's standard terms and conditions for credit products.*

3.16 Information Security for AI Systems

AI systems represent a category of technology infrastructure with specific information security risks, including model theft, adversarial inputs designed to manipulate outputs, and unauthorised access to sensitive training data. The Bank's Technology Department, in conjunction with the AI Governance and Analytics Department, is responsible for implementing and maintaining the following security controls:

- **Role-Based Access Control (RBAC)** — Access to AI model outputs, training data, model parameters, and audit logs is restricted on a need-to-know basis, aligned with each staff member's credit authority tier and functional role. Access rights are reviewed annually and upon any change in role.
- **Model Parameter Protection** — Trained model parameters and weights are classified as confidential assets. They are stored in secured environments with access restricted to authorised AI Governance and Analytics staff. Model parameter files are not transmitted via unencrypted channels.
- **Adversarial Input Detection** — The AI Decision Support System includes automated checks for input patterns inconsistent with legitimate credit applications, which may indicate adversarial manipulation attempts. Suspected adversarial inputs trigger an automatic referral to the Compliance Department and the AI Governance and Analytics Department.
- **Penetration Testing** — AI system components, including the Decision Support Dashboard, model APIs, RAG engine, and audit log database, are subject to annual penetration testing conducted by an independent security function. Findings are remediated under the Technology Department's standard vulnerability management process.
- **Supply Chain Security** — Third-party AI technology providers are subject to the Bank's Vendor Risk Management Policy, which requires documented evidence of their own information security standards, data handling practices, and breach notification procedures prior to contract execution.

3.17 Audit Logging and Decision Traceability

Complete and tamper-proof audit logging is a mandatory requirement for every AI-assisted credit decision. The Decision Audit Register is the primary record of AI system activity and human review actions. It supports regulatory examination, internal audit, dispute resolution, and model performance monitoring.

3.17.1 Mandatory Audit Log Fields

The following data fields are recorded in the Decision Audit Register for every credit application processed through the AI Decision Support System:

- Application reference number and date of submission.
- AI model version in operation at time of assessment.
- Input data quality check outcome.
- Computed PD percentage and assigned risk tier.
- XAI module — top five contributing factors and their directional influence.
- RAG retrieval — policy sections retrieved and citation references.
- Reviewing officer identifier and authority tier.
- Decision outcome (Approved / Declined / Escalated / Exception Requested).
- Override flag — Yes/No; if Yes, override direction and rationale summary.
- Decision timestamp and digital signature of authorising officer.

- Credit Assessment Memorandum reference (for Tier 2 and Tier 3 cases).

3.17.2 Log Integrity Controls

Audit log records are write-once at the point of finalisation and cannot be modified without generating a formal Audit Log Amendment Request, approved by the Chief Risk Officer and recorded as an amendment event in the log itself. Log integrity is verified through automated hash-chain validation on a weekly basis. Integrity failures trigger an immediate AI incident report at High severity.

3.18 Model Version Control

All AI models deployed by the Bank are subject to formal version control. The Model Inventory Register maintained by the AI Governance and Analytics Department serves as the authoritative record of model versioning across all systems.

Version control requirements include:

- Every production model version is assigned a unique version identifier in the format [ModelName]-v[Major].[Minor], where a Major version increment denotes a material change to model architecture or feature set, and a Minor version increment denotes recalibration or parameter adjustment without architectural change.
- Retired model versions are archived in a read-only model repository for a minimum of ten years following retirement, to support retrospective audit of historical credit decisions.
- The version of the model active at the time of each credit decision is recorded in the Decision Audit Register and cannot be altered retrospectively.
- Parallel deployment of multiple model versions in the production environment is prohibited. Only one version of each AI system may be in production use at any time. Version transitions must follow the Change Control process.
- Version control documentation for each model version includes: the Validation Report, Model Approval Certificate, deployment date, known issues at deployment, and all post-deployment monitoring reports produced during the version's operational period.

Policy Note: *Version control records are retained regardless of whether the model was ultimately determined to have performed satisfactorily. Records of underperforming model versions are of particular regulatory and audit value and must be preserved with the same rigour as records of successful deployments.*

3.19 Employee Responsibilities in AI Governance

AI governance is a shared institutional responsibility. The specific obligations of each role are defined below. Ignorance of these obligations is not a defence against a finding of non-compliance.

Role	AI-Related Responsibilities	Prohibited Actions
Loan Officer	Review AI output; complete review checklist; document assessment; escalate inconsistencies	Approve/decline solely on AI output; share AI outputs externally
Senior Credit Analyst	Prepare assessment memorandum; evaluate AI-qualitative consistency; document overrides	Submit memorandum without AI output review; suppress override rationale
Credit Manager	Authorise commitment decisions; countersign overrides; review exception applications	Issue credit commitment without complete file review; bypass HITL requirements
Chief Credit Officer	Approve exceptions; review exception register; report to Board Credit Committee	Authorise systematic policy deviations without CRO alignment
CRO	Model governance oversight; AI incident escalation; Board risk reporting	Approve models without independent validation; suppress material model risk findings
AI Governance & Analytics	Model lifecycle management; performance monitoring; fairness audits; RAG stewardship	Deploy model changes without governance committee approval
Internal Audit	Independent assurance on AI controls; HITL compliance testing; audit log review	Participate in model development or deployment decisions

3.19.1 Mandatory AI Training Requirements

All staff whose roles involve interaction with AI system outputs must complete the following training requirements before being granted access to the AI Decision Support System:

- Foundation AI Literacy Programme — Covers the nature of AI models, the AI Decision Lifecycle, HITL obligations, and the governance principles of this chapter. Required for all Loan Officers, Credit Analysts, and Credit Managers.
- XAI Interpretation Training — Covers the practical interpretation of Explainable AI output, the identification of inconsistencies, and the documentation obligations for AI output review. Required for all Credit Officers and Senior Credit Analysts.
- AI Governance Certification — Annual recertification required for all credit authority holders, covering updates to this chapter, model performance developments, and relevant regulatory changes.
- Incident Reporting Training — Covers the identification and reporting of AI incidents, required for all staff interacting with AI systems.

3.20 AI Risk Management

AI risk is managed within the Bank's Enterprise Risk Management framework established in Chapter 1 and further defined in Chapter 2. The following AI-specific risk register identifies the material risk

categories applicable to the Bank's current AI deployment, together with their primary controls and residual risk owners:

AI Risk	Likelihood	Impact	Primary Control	Residual Risk Owner
Model overfitting — poor generalisation to new data	Medium	High	Regular validation; PSI monitoring; retraining policy	CRO
Data poisoning — corrupted training data	Low	Critical	Data lineage controls; source verification; access restrictions	CRO + Technology
Algorithmic bias — disparate impact on borrower groups	Medium	High	Quarterly fairness audit; demographic parity testing	CRO + Compliance
Automation bias — over-reliance on AI by officers	Medium	High	HITL policy enforcement; override culture; training	Chief Credit Officer
XAI failure — explanations inconsistent or unintelligible	Low	Medium	XAI testing; explanation quality review	AI Governance & Analytics
RAG hallucination — incorrect policy retrieval	Low–Medium	High	Officer validation obligation; retrieval accuracy testing	AI Governance & Analytics
Model retirement failure — continued use of stale model	Low	High	Model Inventory Register; mandatory revalidation triggers	CRO
Unauthorised model access	Low	Critical	Role-based access control; audit logging; penetration testing	Technology + Compliance

The AI Risk Register is reviewed quarterly by the AI Governance and Analytics Department and reported to the Chief Risk Officer. Material changes to the risk profile — including the identification of new risk categories or a change in the assessment of existing risks — are reported to the Model Governance Committee and the Board Risk Committee within the next scheduled reporting cycle.

3.21 AI Ethics

The Bank's AI Ethics framework governs the values and constraints that apply to all AI systems deployed in a credit context. AI ethics are not aspirational — they are operational requirements embedded in the design, governance, and monitoring of every AI system the Bank operates.

3.21.1 Ethical Red Lines

The following AI practices are absolutely prohibited within Stratum Capital Bank, irrespective of commercial rationale, technological feasibility, or regulatory ambiguity:

- The use of any characteristic constituting a protected ground under applicable non-discrimination principles — including gender, age, ethnicity, national origin, religion, disability status, or marital status — as an input variable or feature proxy in any credit-related AI model.
- The deployment of any AI system that produces binding credit decisions without human review and authorisation, including any system that communicates an approval or decline to an applicant without officer sign-off.
- The use of AI systems to profile borrowers for purposes beyond the specific credit assessment context for which the system was validated and approved.
- The concealment of AI-related limitations, performance issues, or bias findings from the Model Governance Committee, the Chief Risk Officer, or the Board Risk Committee.
- The procurement or deployment of any AI system from a third-party provider that does not meet the Bank's Model Transparency and data governance standards, regardless of cost or commercial pressure.

3.21.2 Ethical AI Design Obligations

All AI systems developed for or procured by the Bank must demonstrate the following ethical design attributes before receiving a Model Approval Certificate:

- Purpose Limitation — The model is designed for a specific, documented credit risk purpose and is not capable of being repurposed for non-credit activities without full re-validation and re-approval.
- Reversibility — The Bank can suspend, override, or retire any AI system at any time without creating an operational dependency that cannot be managed through the fallback procedures defined in Section 3.22.
- Proportionality in Automation — The degree of automation applied in the AI decision support process is proportionate to the confidence level in the model's performance and the risk magnitude of the decisions it supports.
- Human Dignity — AI systems interacting with or producing outputs about borrowers must do so in a manner consistent with the Bank's Customer Fairness Principles and the reasonable expectations of dignity and respect that borrowers are entitled to hold.

3.22 Business Continuity for AI Systems

AI system availability is not a prerequisite for the Bank's credit operations. The Bank maintains documented fallback procedures for all AI systems that ensure credit assessment can continue — at a reduced efficiency but acceptable quality level — in the event of AI system unavailability.

Disruption Scenario	Immediate Response	Fallback Procedure	Recovery Target
AI Decision Dashboard unavailable	Notify Technology; log incident	Manual credit assessment using standard paper-based criteria	4 hours RTO
PD model producing erroneous outputs	Suspend model outputs; notify CRO	Senior Credit Analyst manual PD estimation using scorecard	48 hours
RAG Policy Engine unavailable	Notify AI Governance; alert credit staff	Access physical/PDF policy library on shared drive	2 hours RTO
Training data repository corrupted	Isolate corrupted data; notify Technology + CRO	Revert to last validated training dataset	72 hours
AI audit log system failure	Activate manual decision logging protocol	Paper-based decision register; scan to secure storage	1 hour RTO

All fallback procedures are tested at least annually as part of the Bank's Business Continuity Programme. The Technology Department and AI Governance and Analytics Department jointly own the AI system BCP documentation. Training in fallback procedures is a component of the AI Governance Certification required of all credit authority holders.

Policy Note: The existence of fallback procedures does not diminish the Bank's obligation to restore AI system availability within the defined Recovery Time Objectives. Fallback procedures are temporary measures — not an acceptable long-term operational state. Prolonged reliance on fallback procedures beyond the RTO triggers escalation to the Chief Risk Officer and a formal Technology Incident Report.

3.23 AI Model Lifecycle Governance Summary

The following table consolidates the governance requirements applicable at each stage of the AI model lifecycle, providing a single-reference view for the Model Governance Committee, the Chief Risk Officer, and the Internal Audit function:

Lifecycle Stage	Key Activities	Governance Requirement	Responsible Party
1. Development	Data selection; feature engineering; model training	Development standards compliance; ethics screening	AI Governance & Analytics
2. Validation	Independent testing; performance benchmarking; bias audit	Validation Report; CRO sign-off required	Independent Model Validation Unit

Lifecycle Stage	Key Activities	Governance Requirement	Responsible Party
3. Approval	Model Governance Committee review; documentation sign-off	Formal Model Approval Certificate	Model Governance Committee
4. Deployment	Production release; user training; dashboard integration	Deployment checklist; change control record	AI Governance & Analytics + Technology
5. Monitoring	Performance tracking; drift detection; override analysis	Monthly dashboard; quarterly CRO report	AI Governance & Analytics
6. Retraining	Data refresh; parameter recalibration; re-validation	Retraining Justification Report; full re-approval	AI Governance & Analytics
7. Retirement	Decommission; archive; successor deployment	Retirement Notice; Board communication if material	CRO + Model Governance Committee

3.24 Future AI Strategy

Stratum Capital Bank's future AI strategy is guided by the principle that AI capability development must remain within the bounds of the Bank's governance capacity to oversee, explain, and control the systems it deploys. The Bank will not adopt AI systems — however commercially promising — that cannot satisfy the governance requirements established in this chapter at the time of intended deployment.

3.24.1 Strategic AI Development Priorities

The Bank's forward-looking AI strategy focuses on the following development priorities, each of which will be subject to full governance assessment before any implementation proceeds:

- **Enhanced PD Model Precision** — Investigation of advanced modelling techniques that improve PD estimation accuracy for underserved borrower segments, while maintaining full explainability and compliance with fairness standards. Any technique that reduces model explainability below the standards defined in this chapter will not proceed to deployment.
- **Automated Vulnerability Detection** — Development of supplementary models capable of identifying early indicators of borrower financial vulnerability in the application and portfolio monitoring stages. This capability is intended to enhance, not replace, the Credit Officer's Vulnerability Assessment obligations defined in Chapter 2.
- **RAG System Enhancement** — Expansion of the RAG Policy Retrieval Engine's knowledge base to incorporate additional internal policy libraries, historical exception records, and credit committee precedents, improving the policy guidance available to Credit Officers during application review.
- **Conversational AI Policy Assistant** — Evaluation of a conversational interface layer for the Internal Policy Search Engine, enabling Credit Officers to query the policy library in natural language. This system will operate in an advisory capacity only and will clearly distinguish between authoritative policy text and AI-generated interpretation.
- **Stress Testing Integration** — Development of AI-assisted stress testing capabilities that allow the Credit Risk Management Department to model the portfolio-level impact of defined macroeconomic scenarios on PD distributions across borrower segments and product types.

3.24.2 AI Governance Evolution

The Bank's AI Governance Framework will be reviewed and updated at each Annual Review cycle to reflect:

- Developments in AI regulation applicable to financial services institutions.
- Advances in AI explainability and fairness assessment methodologies.
- Operational lessons arising from AI incident management, model performance monitoring, and Internal Audit findings.
- Recommendations from the Model Governance Committee arising from its quarterly review activities.
- Feedback from Credit Officers regarding the practical usability and reliability of AI system outputs in their day-to-day credit assessment work.

The Chief Risk Officer is responsible for ensuring that the governance framework evolves at a pace commensurate with the Bank's AI capability development. The Bank will not deploy AI capabilities that outpace the governance framework's capacity to provide adequate oversight.

Policy Note: *The Bank's future AI strategy is a living document, reviewed annually by the Executive Committee and presented to the Board Technology and Risk Committee. Strategic AI initiatives that require new governance structures or that represent a material departure from the AI philosophy established in this chapter require Board approval before development commences.*

3.25 Chapter Summary

Chapter 3 of the Stratum Capital Bank Enterprise Governance & Risk Framework has established the comprehensive AI governance structure applicable to all AI systems deployed in the Bank's credit assessment and risk management activities. The following provisions are binding on all relevant staff, functions, and third-party providers:

- The AI Vision establishing trustworthy, explainable, ethical, and governed AI as the four non-negotiable pillars of the Bank's approach to AI deployment.
- Ten AI Governance Principles defining the non-negotiable standards for human authority, explainability, proportionality, independent validation, data integrity, non-discrimination, audit completeness, continuous improvement, and accountability.
- A complete inventory of the five AI systems currently operated by the Bank and their human review requirements.
- The Human-in-the-Loop Decision Framework including the Enterprise Decision Matrix, HITL integrity obligations, and the consequences of HITL non-compliance.
- The nine-stage AI Decision Lifecycle defining the sequence from application submission through audit log finalisation, with the human judgment and human decision steps identified as non-automatable.
- The Explainable AI Policy defining minimum explainability standards — including factor identification, directional clarity, and plain-language output — and the interpretation obligations of Credit Officers.
- Model Transparency requirements and the Model Inventory Register documentation standards.
- The Bias Detection and Fairness Assessment framework covering five fairness dimensions and mandatory remediation procedures for breach findings.
- The Model Validation Policy requiring independent validation, a Validation Report, and a Model Approval Certificate before any model proceeds to production.
- The Model Monitoring Policy defining six performance KPIs, alert thresholds, and the monthly AI Model Performance Dashboard governance cycle.
- The Model Retraining Policy defining five retraining triggers, process requirements, and intensified post-deployment monitoring obligations.
- The AI Incident Management framework covering four severity levels, response matrices, reporting obligations, and post-incident review requirements.
- Data Governance standards covering six data categories, permissible and prohibited uses, and retention periods.
- Data Privacy obligations specific to AI operations, including anonymisation for training, automated decision disclosure, and cross-border data transfer restrictions.
- Information Security controls including role-based access, model parameter protection, adversarial input detection, and supply chain security requirements.
- Audit Logging requirements with eleven mandatory data fields and log integrity controls.
- Model Version Control requirements including unique version identifiers, archival obligations, and parallel deployment prohibition.
- Employee Responsibilities across all credit and governance roles, with prohibited actions defined for each role and mandatory AI training requirements.
- The AI Risk Register covering eight material AI risk categories with likelihood, impact, primary controls, and residual risk owners.
- AI Ethics framework including five ethical red lines and four ethical design obligations for all AI systems.

- Business Continuity fallback procedures for five disruption scenarios with defined Recovery Time Objectives.
- Future AI Strategy priorities and AI Governance Evolution commitments subject to annual Board review.

The provisions of this chapter, taken together with Chapters 1 and 2, establish the complete philosophical, risk management, and AI governance foundation for all subsequent operational policies produced by Stratum Capital Bank. Product-level AI policies, vendor AI governance requirements, and credit operations procedures must be consistent with this chapter and derived from it.

— End of Chapter 3 —

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